Functions & Graphs

ANSWER KEY

Section 1: Function Evaluation & Inverses

1 If $f(x) = 3x^2 - 4x + 7$, what is the value of $f(-2)$?

a. 11

b. 21

c. 27 ✓

d. 31

$f(-2) = 3(-2)^2 - 4(-2) + 7 = 3(4) + 8 + 7 = 12 + 8 + 7 = 27$.

2 Given the linear function $f(x) = \frac{4x-7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+7}{4}$ ✓

b. $f^{-1}(x) = \frac{3x-7}{4}$

c. $f^{-1}(x) = \frac{4x+3}{7}$

d. $f^{-1}(x) = \frac{7x+3}{4}$

Set $y = \frac{4x-7}{3} \Rightarrow 3y = 4x - 7 \Rightarrow 4x = 3y + 7 \Rightarrow x = \frac{3y+7}{4}$. Swapping variables gives $f^{-1}(x) = \frac{3x+7}{4}$.

3 If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$ ✓

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

The expression under the radical must be non-negative: $2x + 10 \geq 0 \Rightarrow 2x \geq -10 \Rightarrow x \geq -5$.

Section 2: Composite Functions & Domain/Range

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4 Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46 ✓

c. 49

d. 52

Evaluate inner function first: $f(1) = 2(1) + 5 = 7$. Then $(g \circ f)(1) = g(7) = 7^2 - 3 = 49 - 3 = 46$.

5 If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$ ✓

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

$h(k(x)) = 3k(x) - 1 = 6x + 8 \Rightarrow 3k(x) = 6x + 9 \Rightarrow k(x) = 2x + 3$.

6 What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$ ✓

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

The parabola opens downward because of the negative leading coefficient. The vertex is at $(4, 9)$, which is a maximum. Thus the range is $y \leq 9$.

Section 3: Graph Transformations & Symmetries

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7 The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$ ✓

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

A horizontal shift to the right by h replaces x with $x - h$. A downward shift of k subtracts k from the function. Thus $y = f(x - 3) - 5$.

8 Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$ ✓

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

Check $f(x) = 4x^2 + 7$: $f(-x) = 4(-x)^2 + 7 = 4x^2 + 7 = f(x)$. Therefore, it is even.

9 What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$ ✓

c. $y = 2$

d. No horizontal asymptote

Since degrees of numerator and denominator are equal (both 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{6}{2} = 3$.

Section 4: Piecewise & Special Functions

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10 A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7 ✓

b. 8

c. 9

d. 11

For $x = -2 < 1$: $f(-2) = 2(-2) + 3 = -1$. For $x = 3 \geq 1$: $f(3) = 3^2 - 1 = 8$. Sum $= -1 + 8 = 7$.

11 If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6 ✓

c. 7

d. 1

$[3.8] = 3$ and $[-2.4] = -3$ (the greatest integer less than or equal to -2.4). Therefore, $3 - (-3) = 6$.

12 For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3 ✓

c. 4

d. 5

For continuity at $x = 2$, the left limit must equal the right limit:
$c(2) + 4 = 3(2)^2 - 2 \Rightarrow 2c + 4 = 12 - 2 = 10 \Rightarrow 2c = 6 \Rightarrow c = 3$.

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